DECENT COACHING CLASSES

IX CBSE

Mind Your Math - IX

Time: 3 Hours **Maximum Marks: 80**

General Instructions:

1. This question paper contains **38** questions. **All** questions are compulsory.
2. The question paper is divided into **five sections** – A, B, C, D and E.
3. **Section A** has 20 Multiple Choice Questions (Q.1–Q.20) of 1 mark each.
4. **Section B** has 5 Very Short Answer questions (Q.21–Q.25) of 2 marks each.
5. **Section C** has 6 Short Answer questions (Q.26–Q.31) of 3 marks each.
6. **Section D** has 4 Long Answer questions (Q.32–Q.35) of 5 marks each.
7. **Section E** has 3 Case Based questions (Q.36–Q.38) of 4 marks each.
8. Internal choice is provided in Q.27, Q.33 and Q.35. Only one of the alternatives is to be attempted.
9. Graph paper is required for Q.32. Use of calculator is **not** allowed.

SECTION A (20 × 1 = 20 marks)

Select the correct option in each of the following.

1. The point (−3, 4) is reflected in the y-axis and the image so obtained is then reflected in the x-axis. The final point is

(a) (3, −4) (b) (−3, −4) (c) (3, 4) (d) (−3, 4)

2. If the distance between the points (x, 0) and (0, 4) is 5 units, then x =

(a) 3 (b) −3 (c) ±3 (d) ±5

3. A linear polynomial p(x) = ax + b has zero −3 and p(1) = 8. Then (a, b) =

(a) (2, 6) (b) (6, 2) (c) (4, 12) (d) (1, 3)

4. The graph of y = 2x − 4 meets the x-axis at A and the y-axis at B. The area of triangle OAB (O is the origin) is

(a) 2 sq. units (b) 4 sq. units (c) 6 sq. units (d) 8 sq. units

5. Which of the following numbers is irrational?

(a) $\sqrt{12}/\sqrt{3}$ (b) $(\sqrt{5}+\sqrt{2})(\sqrt{5}-\sqrt{2})$
(c) $(2+\sqrt{3})^2$ (d) $\sqrt{0.09}$

6. The value of $81^{-3/4} \times (1/9)^{-1}$ is

(a) 1/3 (b) 3 (c) 1/27 (d) 27

7. Which of the following numbers lies between 1/3 and 1/2 and is irrational?

(a) 0.4 (b) 0.444…
(c) 0.404004000400004… (d) 0.45

8. The decimal expansion of 7/80 terminates after how many decimal places?

(a) 2 (b) 3 (c) 4 (d) 5

9. If $a - 1/a = 4$, then $a^2 + 1/a^2$ =

(a) 14 (b) 16 (c) 18 (d) 20

10. $(x+y)^3 - (x-y)^3$ =

(a) $2y^3$ (b) $6x^2y$ (c) $6x^2y + 2y^3$ (d) $2x^3 + 6xy^2$

11. If $x^2 + kx + 36$ is a perfect square, then k =

(a) 6 (b) 12 (c) ±6 (d) ±12

12. A chord of length 16 cm is at a distance of 6 cm from the centre of a circle. The radius of the circle is

(a) 8 cm (b) 10 cm (c) 12 cm (d) 14 cm

13. ABCD is a cyclic quadrilateral in which $\angle A = (2x + 4)°$ and $\angle C = (4x - 4)°$. The value of x is

(a) 20 (b) 25 (c) 30 (d) 35

14. An arc of a circle subtends an angle of 140° at the centre. The angle subtended by this arc at a point on the major arc is

(a) 35° (b) 70° (c) 140° (d) 280°

15. The area of a triangle with sides 13 cm, 14 cm and 15 cm is

(a) 84 cm^2 (b) 91 cm^2 (c) 105 cm^2 (d) 42 cm^2

16. The perimeter of an equilateral triangle is 60 m. Its area is

(a) $100\sqrt{3}$ m^2 (b) $200\sqrt{3}$ m^2 (c) $50\sqrt{3}$ m^2 (d) $400\sqrt{3}$ m^2

17. A die is thrown once. The probability of getting a prime number or an even number is

(a) 1/2 (b) 2/3 (c) 5/6 (d) 1

18. A bag contains 6 red balls and some blue balls. If the probability of drawing a blue ball is 2/5, the number of blue balls is

(a) 3 (b) 4 (c) 5 (d) 6

19. The 5th term of an AP is 22 and its 9th term is 38. The 15th term is

(a) 58 (b) 60 (c) 62 (d) 64

20. The sum of the first n terms of a sequence is $S_n = 2n^2 + 3n$. Its 10th term is

(a) 39 (b) 41 (c) 43 (d) 45

SECTION B (5 × 2 = 10 marks)

21. Show that the points A(−2, 1), B(2, 1) and C(0, 5) are the vertices of an isosceles triangle. **[2]**

22. The graph of the linear equation $y = mx + c$ passes through the points (1, 5) and (3, 11). Find the values of m and c. **[2]**

23. Express 0.3666… (the digit 6 repeating) in the form p/q, where p and q are integers and $q \neq 0$. **[2]**

24. If $x + y = 5$ and $xy = 6$, find the value of $x^3 + y^3$. **[2]**

25. Prove that a parallelogram inscribed in a circle is a rectangle. **[2]**

SECTION C (6 × 3 = 18 marks)

26. Show that the points A(1, 2), B(5, 4), C(3, 8) and D(−1, 6) are the vertices of a square. **[3]**

27. If $(3 + \sqrt{2})/(3 - \sqrt{2}) = a + b\sqrt{2}$, where a and b are rational numbers, find a and b. **[3]**

OR

Simplify: $(7 + 3\sqrt{5})/(3 + \sqrt{5}) - (7 - 3\sqrt{5})/(3 - \sqrt{5})$

28. Factorise: $8x^3 + 12x^2 + 6x - 7$ **[3]**

29. The sides of a triangular park are in the ratio 5 : 12 : 13 and its perimeter is 300 m. Find the area of the park using Heron's formula. Also find the cost of levelling it at ₹ 4 per m^2. **[3]**

30. Cards numbered 1 to 30 are put in a bag and mixed well. One card is drawn at random. Find the probability that the number on the card is **[3]**

(i) a prime number [1]

(ii) a perfect square [1]

(iii) divisible by both 3 and 5 [1]

31. How many terms of the AP 24, 21, 18, … must be taken so that their sum is 78? Explain why you get two answers. **[3]**

SECTION D (4 × 5 = 20 marks)

32. Draw the graph of the linear equation $2x + 3y = 12$ on graph paper (take 1 cm = 1 unit on both axes). From the graph, find **[5]**

(i) the value of y when x = 3 [1]

(ii) the value of x when y = 6 [1]

(iii) the area of the triangle formed by the line and the coordinate axes. [1]

33. If $x = 1/(3 - 2\sqrt{2})$, find the value of (i) $x + 1/x$ (ii) $x^2 + 1/x^2$ (iii) $x^3 + 1/x^3$. **[5]**

OR

If $a + b + c = 9$ and $ab + bc + ca = 26$, find $a^2 + b^2 + c^2$ and $a^3 + b^3 + c^3 - 3abc$. Hence, if $abc = 24$, find $a^3 + b^3 + c^3$.

34. Two circles with centres O and O′ and radii 5 cm and 3 cm respectively intersect at two points A and B. If OO′ = 4 cm, prove that OO′ is the perpendicular bisector of AB and hence find the length of the common chord AB. **[5]**

35. A field is in the shape of a quadrilateral ABCD in which AB = 9 m, BC = 40 m, CD = 15 m, DA = 28 m and ∠B = 90°. Find the area of the field. **[5]**

OR

The sides of a triangular field are 52 m, 56 m and 60 m. Find its area. Also find the altitude on the side of length 56 m and the length of the shortest altitude of the triangle.

SECTION E (3 × 4 = 12 marks) Case Based Questions

36. **Case Study 1: School Map.** The ground plan of a school is drawn on a coordinate grid, where 1 unit represents 10 m. The main gate is at A(1, 2), the library is at B(7, 2) and the canteen is at C(7, 10). A garden is to be developed at D so that ABCD becomes a rectangle. **[4]**

(i) Find the actual distance between the main gate and the library. [1]

(ii) Find the actual distance between the main gate and the canteen. [1]

(iii) Find the coordinates of the midpoint of AC. [1]

(iv) Find the coordinates of D. [1]

37. **Case Study 2: Savings.** Meera starts saving money for a school trip. She saves ₹ 50 in the first week and, every week after that, she saves ₹ 20 more than the previous week. **[4]**

(i) How much does she save in the 12th week? [1]

(ii) In which week will she save ₹ 410? [1]

(iii) Find the total amount saved by her in the first 15 weeks. [2]

38. **Case Study 3: Spinner Game.** At a school fair, a spinner is divided into 8 equal sectors numbered 1 to 8. Sectors 1, 2 and 3 are red, sectors 4 and 5 are blue, sectors 6 and 7 are green and sector 8 is yellow. The spinner is spun once and the number in the sector where the pointer stops is noted. **[4]**

(i) Find the probability that the pointer stops on a red sector. [1]

(ii) Find the probability that the pointer does not stop on a green sector. [1]

(iii) Find the probability that the number obtained is even. [1]

(iv) Find the probability that the number obtained is a prime number or the sector is blue. [1]

— End of Question Paper —